GSSS INSTITUTE OF ENGINEERING & TECHNOLOGY FOR WOMEN, MYSURU
Department of Electronics & Communication Engineering | GATE Preparation Library

Control Systems — Official GATE ECE Syllabus

Paper Code: EC • Section 6: Control Systems (8 - 10 Marks)

Weightage: 8 - 10 Marks • **Subject Code:** CS • **Key Focus:** Time Response, Root Locus, Bode & Nyquist, State Space

OFFICIAL GATE EXAMINATION PATTERN & MARKING SCHEME
• **Question Types:** Multiple Choice Questions (MCQ, 1/3 negative marks), Multiple Select Questions (MSQ, zero negative marks), Numerical Answer Type (NAT, zero negative marks).
• **Weightage:** General Aptitude (15 Marks) + Engineering Mathematics (13 Marks) + Core Technical Subjects (72 Marks) = **100 Total Marks (65 Questions, 180 Minutes)**.

SECTION: BASIC CONTROL SYSTEM CONCEPTS & MODELING

- Open-loop vs. closed-loop control systems: Feedback effects (sensitivity reduction, bandwidth expansion, stabilization).
- Transfer function concept: Definition (Laplace transform of impulse response with zero initial conditions), poles, zeros, proper and strictly proper transfer functions.
- Block diagram representation and reduction rules: Moving summing points, moving takeoff points, parallel and feedback blocks.
- Signal Flow Graphs (SFG): Nodes, forward paths, feedback loops, non-touching loops.
- Mason's Gain Formula: T = (1 / Delta) * sum(P_k * Delta_k). Systematic determination of forward path gains and graph determinants.
- Modeling of physical systems: Electrical networks, translational and rotational mechanical systems, force-voltage and force-current analogous systems.

SECTION: TIME-DOMAIN ANALYSIS & TRANSIENT RESPONSE

- Standard test signals: Unit step, unit ramp, unit parabolic, unit impulse.
- First-order system response to step, impulse, and ramp inputs: Time constant (tau), settling time.
- Second-order system transient response: Standard characteristic equation s^2 + 2*zeta*omega_n*s + omega_n^2 = 0.
- Damping regimes: Undamped (zeta = 0, purely imaginary poles), Underdamped (0 < zeta < 1, complex conjugate poles), Critically damped (zeta = 1, real and equal poles), Overdamped (zeta > 1, real and distinct poles).
- Transient response specifications for underdamped second-order systems: Delay time (td), Rise time (tr = (pi - beta) / omega_d), Peak time (tp = pi / omega_d where omega_d = omega_n*sqrt(1 - zeta^2)), Maximum Peak Overshoot (Mp = exp(-pi*zeta / sqrt(1 - zeta^2)) * 100%), Settling time (ts = 4 / (zeta*omega_n) for 2% tolerance band, ts = 3 / (zeta*omega_n) for 5% band).
- Steady-state error analysis: System type (number of open-loop poles at origin s=0), static error constants (Position error constant Kp = lim_{s->0} G(s)H(s), Velocity error constant Kv = lim_{s->0} s*G(s)H(s), Acceleration error constant Ka = lim_{s->0} s^2*G(s)H(s)).
- Steady-state errors for step, ramp, and parabolic inputs as a function of system type.

SECTION: STABILITY & ROUTH-HURWITZ CRITERION

- Concept of stability in linear systems: Absolute stability vs. relative stability, BIBO stability criteria.
- Location of poles in s-plane: Left half s-plane (stable), Right half s-plane (unstable), jw-axis poles (marginally stable if simple, unstable if multiple).
- Routh-Hurwitz Stability Criterion: Necessary condition (all coefficients present and same sign), sufficient condition (first column of Routh array contains no sign changes).
- Number of sign changes in the first column equals the number of right-half s-plane roots.
- Special cases in Routh array: First element of a row is zero (epsilon method or reciprocal polynomial substitution), an entire row of zeros (indicates roots symmetric about origin; auxiliary equation A(s) formulation and derivative dA/ds substitution).

SECTION: ROOT LOCUS TECHNIQUE

- Root locus concept: Trajectory of closed-loop poles as open-loop gain K varies from 0 to infinity.
- Angle condition: sum(angle of zeros) - sum(angle of poles) = +-(2q + 1)*180 degrees (determines whether a point in s-plane lies on root locus).
- Magnitude condition: |G(s)H(s)| = 1 (determines the value of K for a valid point on the root locus).
- Construction rules: Number of branches = max(P, Z), symmetry about real axis, real-axis segments (root locus exists where total number of real poles and zeros to the right is odd).
- Asymptotes: Number of asymptotes = P - Z, angles of asymptotes phi_A = (2q + 1)*180 / (P - Z), centroid of asymptotes sigma_A = (sum(Real parts of poles) - sum(Real parts of zeros)) / (P - Z).
- Breakaway and break-in points: Derived from dK/ds = 0 (valid only if point lies on root locus segment).
- Angle of departure from complex poles and angle of arrival at complex zeros.
- Points of intersection with imaginary (jw) axis: Solved using Routh array auxiliary equation or by setting s = jw.

SECTION: FREQUENCY-DOMAIN ANALYSIS (BODE & NYQUIST)

- Frequency response specifications: Resonant peak (Mr = 1 / (2*zeta*sqrt(1 - zeta^2))), Resonant frequency (omega_r = omega_n*sqrt(1 - 2*zeta^2)), Bandwidth (BW = omega_n * sqrt(1 - 2*zeta^2 + sqrt(4*zeta^4 - 4*zeta^2 + 2))).
- Bode plot: Semi-logarithmic magnitude plot (dB vs. log omega) and phase plot (degrees vs. log omega).
- Corner frequencies, slope contributions (+-20 dB/dec for simple zeros/poles, +-40 dB/dec for complex pairs).
- Gain Margin (GM = -20*log10|G(j*omega_pc)|) at phase crossover frequency (angle = -180 deg).
- Phase Margin (PM = 180 + angle(G(j*omega_gc))) at gain crossover frequency (|G(j*omega_gc)| = 1).
- Stability criteria from Bode plots: For minimum-phase systems, stable if omega_gc < omega_pc (GM > 0 dB and PM > 0 deg).
- Polar plots: Direct sketching of magnitude and phase as omega goes from 0 to infinity.
- Nyquist Stability Criterion: Principle of Argument (N = P - Z where N = counter-clockwise encirclements of -1+j0, P = open-loop poles in RHP, Z = closed-loop poles in RHP).
- Nyquist contour mapping, indentation around poles at the origin, relative stability from Nyquist diagram.

SECTION: COMPENSATORS & STATE-SPACE ANALYSIS

- Compensator design: Lead compensator (dominant zero, adds positive phase angle, increases bandwidth, improves transient speed, increases PM), Lag compensator (dominant pole, improves steady-state accuracy by increasing low-frequency gain, reduces bandwidth), Lag-Lead compensator.
- Maximum phase lead angle: sin(phi_m) = (1 - alpha) / (1 + alpha) at frequency omega_m = 1 / (T*sqrt(alpha)).
- State-space representation: State variables, state vector x(t), input vector u(t), output vector y(t). State equations: dx/dt = A*x + B*u, y = C*x + D*u.
- State Transition Matrix (STM): phi(t) = exp(A*t) = L^-1{(sI - A)^-1}. Properties: phi(0) = I, phi(t1+t2) = phi(t1)*phi(t2), [phi(t)]^-1 = phi(-t).
- Solution of state equations: Zero-input response + Zero-state response (convolution with input).
- Controllability: Kalman's controllability matrix Qc = [B | AB | A^2*B | ... | A^(n-1)*B]. System is controllable if rank(Qc) = n (det(Qc) != 0).
- Observability: Kalman's observability matrix Qo = [C^T | A^T*C^T | ... | (A^T)^(n-1)*C^T]. System is observable if rank(Qo) = n (det(Qo) != 0).
- State feedback controller design: Pole placement using Ackermann's formula.

RECOMMENDED STANDARD TEXTBOOKS & REFERENCES

- Modern Control Engineering — Katsuhiko Ogata (Pearson)
- Control Systems Engineering — Norman S. Nise (Wiley)
- Automatic Control Systems — Benjamin C. Kuo & Farid Golnaraghi

HIGH-YIELD GATE EXAM STRATEGY & REVISION TIPS

- A system is stable from Nyquist plot if and only if Z = P - N = 0 (closed-loop RHP poles Z must be zero).
- Peak overshoot Mp depends ONLY on damping ratio zeta; it is completely independent of natural frequency omega_n.
- Transfer function from state-space matrices: T(s) = C * (sI - A)^-1 * B + D.